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Computer Viruses and Antivirus Software

1 Introduction: What Are Computer Viruses?

A **computer virus** is a type of malicious software (malware) that spreads from one computer to another and causes harm to files, programs, or the operating system.

A virus usually:

- Attaches itself to a file or program
- Spreads when the infected file is opened
- Deletes or damages data
- Slows down the computer
- Steals information in some cases

Just like a human virus spreads from person to person, a computer virus spreads from device to device.

2 Types of Computer Viruses

➤ File Infector Virus

- Attaches itself to executable files (.exe) and activates when the file is opened.
- **Extra Info:** One of the most common types of virus. Can modify or delete files and sometimes spread to network drives.
- **Example:** CIH Virus (Chernobyl Virus) – damaged Windows files and even the BIOS in 1998.
- **Effect:** Corrupts files, slows down programs, may cause system crashes.

➤ Boot Sector Virus

- Infects the boot sector of a computer, which is the first part loaded during startup.
- **Extra Info:** Loads into memory before the operating system, making it hard to detect.
- **Example:** Brain Virus – one of the first boot sector viruses, appeared in 1986.
- **Effect:** Can prevent the computer from starting, delete files, and corrupt the hard drive.

➤ Macro Virus

- Spreads through documents such as Microsoft Word or Excel files; activates when the document is opened.

- **Extra Info:** Exploits macros (automated commands) inside documents to infect other files. Often used in phishing attacks.
- **Example:** Melissa Virus – infected Word documents in 1999 and spread via email.
- **Effect:** Can delete or copy files, send infected documents to contacts, or slow down the computer.

➤ **Resident Virus**

- Installs itself in the computer's memory (RAM) and stays active even after the infected program is closed.
- **Extra Info:** Continuously monitors files for execution and infects them automatically. Harder to remove because it stays in memory.
- **Example:** Jerusalem Virus – appeared in 1987, slowed systems and deleted files on Friday the 13th.
- **Effect:** Slows system performance, corrupts executable files, causes system instability.

➤ **Direct Action Virus**

- Infects files immediately when executed but does not stay in memory.
- **Extra Info:** Only activates when the infected file runs. Easier to detect and remove compared to resident viruses.
- **Example:** Vienna Virus – one of the early direct action viruses that infected .exe files.
- **Effect:** Modifies or deletes files during execution, but stops spreading when the system restarts.

➤ **Polymorphic Virus**

- Changes its code every time it spreads, making it difficult for antivirus programs to detect.
- **Extra Info:** Uses encryption and mutation techniques to hide from signature-based antivirus software.
- **Example:** Marburg Virus – modified itself to avoid detection in the 1990s.
- **Effect:** Hard to detect, can continue spreading without being noticed, often slows systems or corrupts files.

➤ **Multipartite Virus**

- Infects both the boot sector and executable files, spreading in multiple ways.

- **Extra Info:** Very dangerous because it can reinfect the system even after part of it is removed.
- **Example:** Tequila Virus – attacked DOS systems and spread through files and boot sectors.
- **Effect:** Hard to remove completely, can damage multiple areas of the system, may cause crashes and file loss.

3 What Are Antivirus Software?

Antivirus software is a program designed to detect, prevent, and remove malicious software, including viruses.

It works by:

- Scanning files and programs for known virus patterns
- Monitoring the system for suspicious behavior
- Removing or quarantining infected files
- Providing real-time protection against new threats

Examples of antivirus software include:

- **Avast**
- **McAfee**
- **Kaspersky**
- **Norton**

4 How Antivirus Protects Against Viruses

- **Real-time scanning** – checks files when they are opened, downloaded, or executed.
- **Automatic updates** – ensures the software can detect the latest viruses.
- **Quarantine feature** – isolates infected files to prevent further damage.
- **Full system scans** – periodically scans all files and programs to detect hidden threats.

Using antivirus software together with safe internet habits is the best way to protect your computer.

5 Prevention Tips

To protect your computer from viruses:

- Avoid clicking unknown links
- Do not download files from untrusted sources
- Install and regularly update antivirus software
- Keep your operating system updated
- Back up important files regularly
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Conclusion

Computer viruses are harmful programs that can damage files and systems, while antivirus software is a protective tool to detect and remove them. Understanding both helps us use computers safely and avoid cyber threats.